Université Paris Dauphine – PSL

M2 MATH

Regularity structures for the singular stochastic partial differential equation ϕ_3^4

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Abstract

This master's thesis explores the mathematical foundations of singular stochastic partial differential equations (SPDEs), focusing on the stochastic quantization of the classical ϕ_3^4 model in quantum field theory. Due to the extremely rough nature of the driving space-time white noise, traditional calculus fails because the non-linear term ϕ^3 is mathematically ill-defined. To overcome these singularities, we investigate Martin Hairer's landmark theory of regularity structures. Specifically, we study a recent, cutting-edge "diagram-free" approach that replaces complex combinatorial trees with a more parsimonious index set of multi-indices. By treating the directional Malliavin derivative as a "tangent vector" to the solution manifold, it can be mathematically analyzed as a modeled distribution. This geometric reformulation, in conjunction with a spectral gap inequality, yields robust stochastic estimates and BPHZ renormalization. Our work provides an overview of the theory of regularity strucutures, and detailed reformulation of certain proofs of this modern approach.

Contents

0.1 Integration

Hypothesis :

$$(\int_{B_R} \mathrm{d}x\mathbb{E}^{\frac{2}{q}} \left|(\delta\Pi^- - \mathrm{d}\Gamma^*_x\Pi^-_x)_{\beta r}(x)\right|^q)^{\frac{1}{2}} \lesssim r^{\alpha-2+\frac{D}{2}}(r+R)^{|\beta|-|\alpha|}, \tag{1}$$

$$(\int_{B_R} \mathrm{d}x\mathbb{E}^{\frac{2}{q}} \left|(\mathrm{d}\Gamma^*_x)^\gamma_\beta\right|^q)^{\frac{1}{2}} \lesssim R^{\frac{D}{2}+|\beta|-|\gamma|} \text{ for populated } \gamma \neq \mathrm{pp}, \tag{2}$$

$$\partial^{\mathbf{n}}((\delta\Pi - \mathrm{d}\Gamma^*_x\Pi_x)_\beta)(x) = 0,\ \forall x \in \mathbb{R}^d, \forall\, |\mathbf{n}| < 2+s \tag{3}$$

Goal :

$$\left(\int_{B_R} \mathrm{d}x\mathbb{E}^{\frac{2}{q}} \left|(\delta\Pi - \mathrm{d}\Gamma^*_x\Pi_x)_{\beta r}(x)\right|^q\right)^{\frac{1}{2}} \lesssim r^{\alpha+\frac{D}{2}}(r+R)^{|\beta|-|\alpha|} \tag{4}$$

Lemme 0.1 (Liouville)**.** *Let g be a smooth function, $m \in \mathbb{N}$ an integer, and $x \in \mathbb{R}^n$ a base point. Then there exists at most on couple (u,p), with u a smooth function and p a polynomial, such that :*

$$\begin{cases} L(u) = g - p \\ \deg(p) \leq m-2 \\ \forall\, |\mathbf{n}| \leq m, \partial^{\mathbf{n}}u(x) = 0. \limsup_{|x-y|\to\infty} \frac{u(y)}{|x-y|^\kappa} < \infty, \text{ for } \kappa \in \mathbb{R},\ \kappa < m. \end{cases}$$

Proof. Let $|\mathbf{n}| \leq m-2$.

$$0 = \partial^{\mathbf{n}}(L(u))(x) = \partial^{\mathbf{n}}g(x) - \partial^{\mathbf{n}}p(x),$$

□

so that p is determined by g. Then, let u and $\tilde{u}$ be two candidate and denote $w = u - \tilde{u}$. We have $L(w) = 0$, and $\limsup_{|x-y|\to\infty} \frac{u(y)}{|x-y|^\kappa} < \infty$. By Liouville's theorem, w is a polynomial with degree at most κ. Then $\forall\, |\mathbf{n}| \leq m, \partial^{\mathbf{n}}w(x) = 0$ shows that $w = 0$.

We have $L(\delta\Pi_\beta - (\mathrm{d}\Gamma^*_x\Pi_x)_\beta) = \delta\Pi^-_\beta - (\mathrm{d}\Gamma^*_x\Pi^-_x)_\beta \bmod \mathcal{P}_{<|\beta|+\frac{D}{2}-2}$. But it is not sufficient to caracterize $\delta\Pi_\beta - (\mathrm{d}\Gamma^*_x\Pi_x)_\beta$ in terms of $\delta\Pi^-_\beta - (\mathrm{d}\Gamma^*_x\Pi^-_x)_\beta$, because the growth at infinity of $\delta\Pi_\beta - (\mathrm{d}\Gamma^*_x\Pi_x)_\beta$ is too big. Indeed, $(\mathrm{d}\Gamma^*_x\Pi_x)_\beta = \sum_{|\gamma|<|\beta|+\frac{D}{2}} (\mathrm{d}\Gamma^*_x)^\gamma_\beta \Pi_{x\gamma}$. Now $\Pi_{x\gamma}(y)$ is expected to grow like $|x-y|^\gamma$. So the sum include terms that grow almost like $2+\frac{D}{2}$. In contrast, the vanishing order of $\delta\Pi_\beta - (\mathrm{d}\Gamma^*_x\Pi_x)_\beta$ at x would be $\alpha+\frac{D}{2}(=2+s)$, which is lower by more than 2 (we are in the regime $\alpha < 0$). This is the reason why we truncate the sum at order $2+s$. Furthermore, because $L\Pi_{x\beta} = \Pi^-_{x\beta} \bmod \mathcal{P}_{<|\beta|-2}$ and $L\delta\Pi_\beta = \delta\Pi^-_\beta \bmod \mathcal{P}_{<|\beta|-2}$, we restrict ourselves to $|\beta| < \lceil 2+s \rceil$. As a result, we have the following

$$L(\delta\Pi_\beta - \sum_{|\gamma|<2+s} (\mathrm{d}\Gamma^*_x)^\gamma_\beta \Pi_{x\gamma}) = \delta\Pi^-_\beta - \sum_{|\gamma|<2+s} (\mathrm{d}\Gamma^*_x)^\gamma_\beta \Pi^-_{x\gamma} \bmod \mathcal{P}_{<s} \tag{5}$$

and the hypothesis of the lemma are verified, so $U_x \coloneqq \delta\Pi_\beta - \sum_{|\gamma|<2+s}(\mathrm{d}\Gamma^*_x)^\gamma_\beta\Pi_{x\gamma}$ is completely determined by $F_x \coloneqq \delta\Pi^-_\beta - \sum_{|\gamma|<2+s}(\mathrm{d}\Gamma^*_x)^\gamma_\beta\Pi^-_{x\gamma}$.

Remarque 0.2. The threshold $2+s$ is too strong for $(\mathrm{d}\Gamma^*_x\Pi^-_x)_\beta$, but we will need to deal with the remainder $\sum_{|\gamma|\geq 2+s}(\mathrm{d}\Gamma^*_x)^\gamma_\beta\Pi^-_{x\gamma}$ and see that this threshold makes sense.

We claim that (**??**) implies

$$\left(\int_{B_R} \mathrm{d}x \mathbb{E}^{\frac{2}{q}} \left| (\sum_{|\gamma|\geq 2+s} (\mathrm{d}\Gamma_x^*)_\beta^\gamma \Pi^-_{x\gamma})_r(x) \right|^q \right)^{\frac{1}{2}} \lesssim r^{\alpha+\frac{D}{2}-2}(r+R)^{|\beta|-|\alpha|}, \tag{6}$$

$$\left(\int_{B_R} \mathrm{d}x \mathbb{E}^{\frac{2}{q}} \left| (\sum_{|\gamma|\geq 2+s} (\mathrm{d}\Gamma_x^*)_\beta^\gamma \Pi_{x\gamma})_r(x) \right|^q \right)^{\frac{1}{2}} \lesssim r^{\alpha+\frac{D}{2}}(r+R)^{|\beta|-|\alpha|}. \tag{7}$$

So that by triangle inequality, we can add to our set of hypothesis

$$\left(\int_{B_R} \mathrm{d}x \mathbb{E}^{\frac{2}{q}} |(F_x)_r(x)|^q \right)^{\frac{1}{2}} \lesssim r^{\alpha+\frac{D}{2}-2}(r+R)^{|\beta|-|\alpha|},$$

and it suffices to show

$$\left(\int_{B_R} \mathrm{d}x \mathbb{E}^{\frac{2}{q}} |(U_x)_r(x)|^q \right)^{\frac{1}{2}} \lesssim r^{\alpha+\frac{D}{2}}(r+R)^{|\beta|-|\alpha|}.$$

The strategy now is to take advantage of the unicity of solution to $Lu = F_x \bmod \mathcal{P}_{<s}$, to find another expression of U_x, that verifies (**??**). The germ (F_x) is $(2\alpha, \alpha+\frac{D}{2}-2)$-coherent and $\alpha+\frac{D}{2}-2$ - homogeneous. Let K be the heat kernel, which is assumed to be a 2-regularising kernel (we would need to truncate it, for it to be compactly supported, but it is easy to deal with the smooth remainder). Define $\tilde{U_x} \coloneqq \mathcal{K}(F)_x = K * (F_x) - T_x^{2+s}(K * (F_x))$. By the Schauder estimate theorem for coherent germs (Theorem 3.17 of [**BCZ24**]), $\tilde{U_x}$ is a well-defined $(2\alpha+2, \alpha+\frac{D}{2})$-coherent and $\alpha+\frac{D}{2}$-homogeneous germ. Plus its homogeneity norm is dominated by the sum of the homogeneity and coherence norm of (F_x). This means that

$$\left(\int_{B_R} \mathrm{d}x \mathbb{E}^{\frac{2}{q}} \left|(\tilde{U_x})_r(x)\right|^q \right)^{\frac{1}{2}} \lesssim r^{\alpha+\frac{D}{2}}((C+R)^{|\beta|-\alpha} + (C+R)^{|\beta|-3\alpha-2}) \tag{8}$$

$$\lesssim r^{\alpha+\frac{D}{2}}(C+R)^{|\beta|-\alpha} (\text{because } 3\alpha+2>\alpha) \tag{9}$$

which is exactly the estimate we want to prove for (U_x).

Let's show that $\tilde{U_x} = U_x$. It suffices to check that both $\tilde{U_x}$ and U_x verifies Liouville's lemma's hypothesis. First, $L\tilde{U_x} = L(K * (F_x)) - L(T_x^{2+s}(K * (F_x))) = F_x +$ polynomial of degree $<$ s. Then,

- The jet condition is verified by construction of $\tilde{U_x}$.
- With (**??**), we claimed that the remainder $\sum_{|\gamma|\geq 2+s}(\mathrm{d}\Gamma_x^*)_\beta^\gamma \Pi_{x\gamma}$ has vanishing order $\alpha+\frac{D}{2} = 2+s$ at x, and together with (**??**), it shows that the jet condition is also verified for U_x.
- We already dealt with the growth condition of U_x, it was the reason we truncated.
- Growth condition for $\tilde{U}$?

which concludes.

It remains to prove that (**??**) implies (**??**) and (**??**). This is done purely by induction.

The sums $\sum_{|\gamma|\geq 2+s}(\mathrm{d}\Gamma_x^*)_\beta^\gamma \Pi_{x\gamma}$ and $\sum_{|\gamma|\geq 2+s}(\mathrm{d}\Gamma_x^*)_\beta^\gamma \Pi^-_{x\gamma}$ is restricted to populated γ that are not purely polynomial : else $\Pi^-_{x\gamma} = 0$, and $(\mathrm{d}\Gamma_x^*)_\beta^\gamma = (\mathrm{d}\Gamma_x^*)_\beta^{\delta_{\mathbf{n}}} = 0$ because $|\mathbf{n}| \geq 2+s$. Furthermore, $|\gamma|_\prec < |\beta|_\prec$ so the induction hypothesis on $\Pi^-_{x\gamma}$ and $\Pi_{x\gamma}$ are available.

$$\left(\int_{B_R} \mathrm{d}x \mathbb{E}^{\frac{2}{q}} \left| (\sum_{|\gamma|\geq 2+s} (\mathrm{d}\Gamma^*_x)^\gamma_\beta \Pi^-_{x\gamma})_r(x) \right|^q \right)^{\frac{1}{2}} \leq \sum_\gamma \left(\int_{B_R} \mathrm{d}x \mathbb{E}^{\frac{2}{q}} \left| ((\mathrm{d}\Gamma^*_x)^\gamma_\beta \Pi^-_{x\gamma})_r(x) \right|^q \right)^{\frac{1}{2}} \text{(triangle inequality)}$$

$$\leq \sum_\gamma \left(\int_{B_R} \mathrm{d}x \mathbb{E}^{\frac{2}{2q}} \left| (\mathrm{d}\Gamma^*_x)^\gamma_\beta \right|^{2q} \right)^{\frac{1}{2}} \left(\int_{B_R} \mathrm{d}x \mathbb{E}^{\frac{2}{2q}} \left| (\Pi^-_{x\gamma})_r(x) \right|^q \right)^{\frac{1}{2}} \text{(Höld}$$

$$\leq \sum_{2+s\leq|\gamma|\leq|\beta|+\frac{D}{2}} R^{\frac{D}{2}+|\beta|-|\gamma|} r^{\gamma-2}$$

$$\lesssim r^{\alpha-2+\frac{D}{2}} (r+R)^{|\beta|-\alpha}.$$

And we proceed the same way for (**??**).

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